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ELECTRONIC DEVICE AND METHOD FOR ADJUSTING DISPLAY REGION BASED ON MOTION

Abstract

An electronic device and a method for adjusting a display region based on motion are disclosed. The method includes: obtaining a first image including a first target object; detecting the first image to obtain a joint of the first target object; calculating a first score according to a first displacement of the first joint; in response to the first score being greater than a threshold, generating an adjustment command for adjusting a first display region of the first image according to the first joint; and outputting the adjustment command.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of Taiwan application serial no. 113107047, filed on Feb. 27, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] This disclosure relates to an image processing technology, and in particular to an electronic device and a method for adjusting a display region based on motion.

Description of Related Art

[0003] Currently, there are many fitness apps that provide demonstration videos for users. Users can use their cell phones to play the demonstration videos and follow the movements of the trainers in the demonstration videos. However, the small screen size of a cell phone often prevents the user from seeing clearly the movements demonstrated by the trainers in the video. Although users can zoom in on localized images by operating the phone, frequent operation of the phone may affect exercise experience of the user.

SUMMARY

[0004] The disclosure provides an electronic device and a method for adjusting a display region based on motion, capable of automatically adjusting a display region of a played image.

[0005] The electronic device for adjusting the display region based on motion of the disclosure includes a processor and a transceiver. The transceiver obtains a first image including a first target object. The processor is coupled to the transceiver and configured to: detect the first image to obtain a first joint of the first target object; calculate a first score according to a first displacement of the first joint; in response to the first score being greater than a threshold, generate an adjustment command for adjusting a first display region of the first image according to the first joint; and output the adjustment command through the transceiver.

[0006] In an embodiment of the disclosure, the processor is further configured to: calculate a second score according to a second displacement of a second joint of the first target object; and in response to the second score being greater than the threshold, generate the adjustment command according to the first joint and the second joint.

[0007] In an embodiment of the disclosure, the processor is further configured to: divide a region of interest of the first image to obtain multiple sub-regions, in which the sub-regions include a first sub-region and a second sub-region; determine that the first joint is located in the first sub-region and the second joint is located in the second sub-region according to the first image; and in response to the second sub-region being adjacent to the first sub-region, generate the adjustment command according to the first joint and the second joint.

[0008] In an embodiment of the disclosure, the sub-regions include a third sub-region corresponding to a third joint of the first target object, in which a third score corresponding to the third joint is greater than the threshold, and the processor is further configured to: in response to the first sub-region being adjacent to the second sub-region and the third sub-region being adjacent to at least one of the first sub-region and the second sub-region, generate the adjustment command according to the first joint, the second joint, and the third joint.

[0009] In an embodiment of the disclosure, the processor is further configured to: generate the first display region according to at least one sub-region including the first sub-region; and in response to

the second sub-region not being adjacent to the at least one sub-region, generate a second display region of the first image according to the second sub-region.

[0010] In an embodiment of the disclosure, the adjustment command instructs that the first display region of the first image is output in a first period and the second display region of the first image is output in a second period.

[0011] In an embodiment of the disclosure, the processor is further configured to: in response to a size of the second display region being different from a size of the first display region, generate the adjustment command including a zoom in operation or a zoom out operation.

[0012] In an embodiment of the disclosure, the processor is further configured to: determine a boundary of the first display region according to a union of the first sub-region and the second sub-region.

[0013] In an embodiment of the disclosure, the processor is further configured to: obtain a first audio file corresponding to the first image through the transceiver; determine a first correlation between the first audio file and the first joint; and calculate the first score according to the first displacement and the first correlation.

[0014] In an embodiment of the disclosure, the processor is further configured to: perform a speech-to-text conversion on the first audio file to generate text; determine a number of words in the text associated with the first joint; and determine the first correlation according to the number of the words.

[0015] In an embodiment of the disclosure, the processor is further configured to: determine a first weight of the first displacement and a second weight of the first correlation according to a category of the first image; and calculate the first score according to the first displacement, the first weight, the first correlation, and the second weight.

[0016] In an embodiment of the disclosure, the processor is further configured to perform: obtain a second image including a second target object according to the transceiver; and generate a playback control command for the first image according to the second image, and output the playback control command through the transceiver.

[0017] In an embodiment of the disclosure, the processor is further configured to: detect the second image to obtain multiple limb joints of the second target object, in which the limb joints include a left elbow joint, a right elbow joint, a left knee joint, and a right knee joint; calculate an angle according to the limb joints, and determine that the angle is within a default range; and in response to the angle being outside the default range, generate the playback control command for pausing the first image.

[0018] In an embodiment of the disclosure, the processor is further configured to: perform facial recognition on a second target object in a second image to obtain multiple joints; calculate an area change of a polygon formed by the joints; and in response to an absolute value of the area change being greater than a change threshold, generate a playback control command for playback speed control.

[0019] In an embodiment of the disclosure, the playback control command is used to reduce a playback speed of the first image.

[0020] In an embodiment of the disclosure, the processor is further configured to: perform facial recognition on a second target object in a second image to obtain multiple joints; calculate an area change of a polygon formed by the joints; and generate a playback control command for playback speed control according to the area change.

[0021] In an embodiment of the disclosure, the processor is further configured to: in response to the area change being greater than a first threshold, generate the playback control command for reducing a playback speed of the first image; and in response to the area change being less than or equal to a second threshold, generate the playback control command for increasing the playback speed of the first image.

[0022] In an embodiment of the disclosure, the processor is further configured to: detect the first

image to obtain a center point and a height of the first target object; and determine a region of interest of the first image according to the center point and the height.

[0023] In an embodiment of the disclosure, the processor is further configured to: output a script through the transceiver, in which the script includes the adjustment command and a timestamp of the first image corresponding to the adjustment command.

[0024] The method for adjusting a display region based on motion of the disclosure includes the following. A first image including a first target object is obtained. The first image is detected to obtain a first joint of the first target object. A first score is calculated according to a first displacement of the first joint. In response to the first score being greater than a threshold, an adjustment command for adjusting a first display region of the first image is generated according to the first joint. The adjustment command is output.

[0025] Based on the above, the electronic device of the disclosure may adjust the display region of the image according to the motion of the person in the image, so that the local details of the image may be more clearly presented to the user.

[0026] To make the aforementioned more comprehensible, several embodiments accompanied with drawings are described in detail as follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate example embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.

[0028] FIG. 1 illustrates a schematic diagram of an electronic device for adjusting a display region based on motion, according to an embodiment of the disclosure.

[0029] FIG. 2 illustrates a schematic diagram of script generation and application according to an embodiment of the disclosure.

[0030] FIG. 3 illustrates a schematic diagram of a region of interest of a demonstration image according to an embodiment of the disclosure.

[0031] FIG. 4 illustrates a schematic diagram of adjustment of a display region according to an embodiment of the disclosure.

[0032] FIG. 5 illustrates a schematic diagram of adjustment of a display region according to an embodiment of the disclosure.

[0033] FIG. 6 illustrates a schematic diagram of a target object according to an embodiment of the disclosure.

[0034] FIG. 7 illustrates a flow chart of a method for adjusting a display region based on motion according to an embodiment of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

[0035] When a user is watching a demonstration image of a trainer for exercise, if the user cannot see the local details of the demonstration image clearly, the user has to manually operate a terminal device to adjust a display region of the demonstration image. The exercise experience of the user will be affected as a result. In order to solve the above problem, the disclosure provides a method for automatically adjusting the display region of an image.

[0036] FIG. 1 illustrates a schematic diagram of an electronic device **100** for adjusting a display region based on motion, according to an embodiment of the disclosure. The electronic device **100** may include a processor **110**, a storage medium **120**, and a transceiver **130**.

[0037] The processor **110** is, for example, a central processing unit (CPU), or other programmable general-purpose or special-purpose micro control unit (MCU), microprocessor, digital signal

processor (DSP), programmable controller, application specific integrated circuit (ASIC), graphics processing unit (GPU), image signal processor (ISP), image processing unit (IPU), arithmetic logic unit (ALU), complex programmable logic device (CPLD), field programmable gate array (FPGA), or other similar components or a combination of the above components. The processor **110** may be coupled to the storage medium **120** and the transceiver **130**, and access and execute multiple modules and various applications stored in the storage medium **120**.

[0038] The storage medium **120** is, for example, any type of fixed or removable random access memory (RAM), read-only memory (ROM), or flash memory, hard disk drive (HDD), solid state drive (SSD), or similar components or a combination of the above components, configured to store multiple modules or various applications that can be executed by the processor **110**.

[0039] The transceiver **130** transmits or receives signals in a wireless or wired manner. The transceiver **130** may also perform, for example, low noise amplification, impedance matching, mixing, up or down frequency conversion, filtering, amplification, and similar operations. The processor **110** may be communicatively connected to an external electronic device through the transceiver **130**.

[0040] FIG. **2** illustrates a schematic diagram of script generation and application according to an embodiment of the disclosure. The electronic device **100** may be communicatively connected to a cloud server **500** and one or more terminal devices (e.g., a terminal device **300** or a terminal device **400**) through the transceiver **130**.

[0041] The user may operate the terminal device **300** to play a demonstration image including a target object **20** (as shown in FIG. **3**), where the target object **20** is, for example, a trainer who instructs the user how to exercise. In addition, the terminal device **300** may capture a user image including a target object **30**, where the target object **30** is, for example, a user who are exercising (i.e., the owner of the terminal device **300**). The terminal device **300** may transmit the user image to the electronic device **100**.

[0042] The electronic device **100** may receive a demonstration image including the target object **20** (e.g., a trainer) through the transceiver **130** (e.g., download a demonstration image from the Internet, such as an aerobic dance demonstration video or a yoga instructional video), or receive a user image including the target object **30** (e.g., the user). The electronic device **100** may generate an adjustment command according to the demonstration image, or may generate a playback control command according to the user image. The electronic device **100** may transmit the adjustment command or the playback control command to the terminal device **300** through the transceiver **130**. The terminal device **300** may adjust the display region of the demonstration image according to the adjustment command, or may control the playback of the demonstration image according to the playback control command.

[0043] In an embodiment, the adjustment command may include, but is not limited to, a zoom in command, a zoom out command, or a pan command.

[0044] In an embodiment, the playback control command may include, but is not limited to a fast forward command, a rewind command, a slow-motion command, a pause command, or a (resume) play command.

[0045] FIG. **3** illustrates a schematic diagram of a region of interest **50** of a demonstration image according to an embodiment of the disclosure. After receiving the demonstration image including the target object **20** (e.g., the trainer), the processor **110** may perform object detection to detect the demonstration image, and then obtain a center point C and a height H of the target object **20** (e.g., the trainer). The processor **110** may determine the region of interest **50** of the demonstration image according to the center point C and the height H of the target object **20** (e.g., the trainer). For example, a shortest distance between an upper boundary (or a lower boundary) of the region of interest **50** and the target object **20** (e.g., the trainer) may be $B*H$, where B may be a default weight. A distance between a left boundary (or a right boundary) of the region of interest **50** and the center point C may be $A*H$, where A may be a default weight.

[0046] The processor **110** may perform the object detection to detect the region of interest **50** of the demonstration image, and then obtain one or more joints of the target object **20** (e.g., the trainer). For example, the processor **110** may obtain a joint **21** corresponding to the right palm of the target object **20** (e.g., the trainer), a joint **22** corresponding to the right elbow of the target object **20** (e.g., the trainer), a joint **23** corresponding to the left palm of the target object **20** (e.g., the trainer), or a joint **24** corresponding to the left foot of the target object **20** (e.g., the trainer). It should be noted that the target object **20** (e.g., the trainer) may have one or more joints that are not shown in FIG. 3 to FIG. 5.

[0047] The processor **110** may calculate a score corresponding to the joint according to a displacement of the joint. In an embodiment, the displacement of a joint may be proportional to the score of the joint. That is, joints with larger displacement may have higher scores.

[0048] In an embodiment, the score of the joint may be associated with an audio file synchronized with the demonstration image. The processor **110** may obtain the audio file synchronized with the demonstration image through the transceiver **130**. The processor **110** may determine a correlation between the audio file and the joint, and may calculate the score of the joint according to the displacement of the joint and the correlation of the joint. Specifically, the processor **110** may perform a speech-to-text (STT) conversion on the audio file to generate text. Next, the processor **110** may determine a number of words in the text associated with the joint, and further determine the correlation between the joint and the audio file according to the number of associated words. For example, if the trainer in the demonstration video keeps repeating the instruction to “turn your right palm”, the processor **110** may determine that the audio file is highly relevant to the joint **21** according to the multiple occurrences of the word “right palm” in the text of the audio file.

[0049] In an embodiment, the storage medium **120** may pre-store a lookup table, in which the lookup table may include mapping relationships between one or more words and default words. Assuming that multiple words having the mapping relationship to the default word appear in the text in the audio file, the processor **110** may determine the correlation between the joint and the audio file according to the number of the words. For example, the lookup table may record the mapping relationship between words such as “buttocks”, “tailbone” or “ischium” and the default word “hip”. If the text in the audio file appears once for “tailbone” once and twice for “ischium”, the processor **110** may determine the correlation between the audio file and the joint corresponding to “buttocks” according to the number “3”.

[0050] In calculating the score of a joint according to the displacement and correlation of the joint, the displacement and correlation may have their respective weights. The processor **110** may calculate the score of the joint according to the displacement, the weight of the displacement, the correlation, and the weight of the correlation. In an embodiment, the processor **110** may determine the weight of the displacement or the weight of the correlation according to a category of the demonstration image. For example, if the theme of the demonstration image is fitness instruction, the trainer may often give instructions to the user in audio in the demonstration image. Accordingly, the processor **110** may increase the weight of the correlation of the joint and reduce the weight of the displacement of the joint to enhance the impact of the audio file on the score calculation of the joint.

[0051] The processor **110** may determine whether the joint can be used to generate an adjustment command for the display region of the demonstration image according to the score of the joint. If the score corresponding to the joint is greater than a threshold, the processor **110** may determine that the joint can be used to generate the adjustment command. If the score corresponding to the joint is less than or equal to the threshold, the processor **110** may determine that the joint cannot be used to generate the adjustment command. For example, if the score of the joint **21** is greater than the threshold and the score of the joint **22** is less than or equal to the threshold, the processor **110** may generate the adjustment command according to the joint **21** rather than the joint **22**. As another example, if the score of the joint **21** is greater than the threshold and the score of the joint **22** is also

greater than the threshold, the processor **110** may generate the adjustment command according to the joint **21** and the joint **22**.

[0052] The processor **110** may divide the region of interest **50** of the demonstration image to obtain multiple sub-regions (e.g., grids), and may determine the sub-regions where each joint is currently located. For example, the processor **110** may determine that the joint **21** is located in a sub-region **51**, the joint **22** is located in a sub-region **52**, the joint **23** is located in a sub-region **53**, or the joint **24** is located in the sub-region **54**. In the following, the sub-region where the joint with score greater than the threshold is located is called a selected sub-region.

[0053] FIG. 4 illustrates a schematic diagram of adjustment of a display region according to an embodiment of the disclosure. The processor **110** may generate a corresponding display region according to the selected sub-regions adjacent to each other. If multiple selected sub-regions are used to generate an adjustment command for a display region, one of the selected sub-regions shall be adjacent to one or more other sub-regions of the selected sub-regions. For example, assuming that the sub-regions **51**, **52**, **53**, and **54** are all selected sub-regions, the processor **110** may determine that the sub-region **51** and the sub-region **52** may be used to generate an adjustment command for a display region **510** based on the sub-region **52** being adjacent to the sub-region **51**, where the display region **510** corresponds to a timestamp (N-1). In addition, the processor **110** may determine that the sub-region **53** can also be used to generate an adjustment command for the display region **510** based on the sub-region **53** being adjacent to the sub-region **51**. Since the sub-region **54** is not adjacent to any of the sub-regions **51**, **52**, or **53**, the processor **110** may determine that the sub-region **54** is not available for generating an adjustment command for the display region **510**.

[0054] In an embodiment, if the selected sub-regions correspond to the display region, the processor **110** may determine a boundary of the display region according to a union of the selected sub-regions. For example, the processor **110** may determine the boundary of the display region **510** according to the union of the selected sub-regions including the sub-regions **51**, **52**, and **53**. Since the sub-region **51** or the sub-region **52** is located at the upper boundary of the union, the processor **110** may determine the upper boundary of the display region **510** according to the sub-region **51** or the sub-region **52**.

[0055] When the timestamp of the demonstration image advances from (N-1) to (N), the processor **110** may generate a new display region **520** for the demonstration image corresponding to the timestamp (N). The selected sub-regions corresponding to the display region **520** include, for example, the sub-regions **51**, **52**, **53**, and **54**. The processor **110** may generate an adjustment command according to the display region **510** corresponding to the timestamp (N-1) and the display region **520** corresponding to the timestamp (N). For example, the processor **110** may generate a reduction command based on a size of the display region **520** being larger than a size of the display region **510**, so that a display of the terminal device **300** may completely display the entire display region **520**. In addition, the processor **110** may generate a pan command based on a center point of the display region **520** being different from a center point of the display region **510**.

[0056] The processor **110** may generate multiple display regions of the demonstration image within the same period, and the selected sub-region of one display region is not adjacent to the selected sub-region of another display region. FIG. 5 illustrates a schematic diagram of adjustment of a display region according to an embodiment of the disclosure. The processor **110** may generate a display region **530** based on multiple selected sub-regions including the sub-regions **51**, **52**, and **53**, and may generate a display region **540** based on multiple selected sub-regions including the sub-region **54**, in which the selected sub-region (e.g., the sub-region **54**) of the display region **540** is not adjacent to the selected sub-region (e.g., the sub-region **51**, **52**, or **53**) of display region **540**.

[0057] The processor **110** may generate an adjustment command according to multiple display regions of the demonstration image. The adjustment command may be used to instruct a terminal device (e.g., the terminal device **300**) to output different display regions at different time periods.

For example, the adjustment command may instruct the terminal device **300** to output the image in the display region **530** during a period (M-1), and to output the image in the display region **540** during a period (M).

[0058] In an embodiment, the processor **110** may generate a pan command based on the display regions of the demonstration image having different center points. For example, the processor **110** may generate a pan command based on a center point of the display region **540** being different from a center point of the display region **530**. The terminal device **300** may pan the output image from the display region **530** to the display region **540** according to the pan command.

[0059] In an embodiment, the processor **110** may generate an adjustment command for a zoom in operation or a zoom out operation based on the display regions of the demonstration image having different sizes. For example, the processor **110** may generate a zoom out command based on a size of the display region **540** being larger than a size of the display region **530**. The terminal device **300** may perform a zoom out operation on the output image according to the zoom out command. Accordingly, the user may view a smaller portion of the demonstration image (with clearer details or higher resolution) at the period (M-1) and a larger portion of the demonstration image (with less clear details or lower resolution) at the period (M).

[0060] The processor **110** may generate a playback control command according to the user image including the target object **30** (i.e., the user of the terminal device **300**). FIG. **6** illustrates a schematic diagram of a target object **30** (e.g., the user) according to an embodiment of the disclosure. In an embodiment, the processor **110** may determine whether the target object **30** (e.g., the user) is still watching the terminal device **300** or has moved away from the terminal device **300** according to the user image, and then determines whether to instruct the terminal device **300** to pause playing the demonstration image. Specifically, the processor **110** may perform object detection on the user image to obtain multiple limb joints of the target object **30** (e.g., the user), in which the limb joints may include a joint **31** representing the left elbow, a joint **32** representing the right elbow, the joint **33** representing the left knee, and the joint **34** representing the right knee. The processor **110** may calculate an angle according to multiple limb joints and determine whether the angle is within a default range. If the angle is within the default range, the processor **110** may determine that the target object **30** (e.g., the user) is still in front of the terminal device **300** (i.e., the target object **30** still appears in the user image captured by the terminal device **300**). Accordingly, the processor **110** may not instruct the terminal device **300** to pause playing the demonstration image. If the angle is outside the default range, the processor **110** may determine that the target object **30** has moved away from the terminal device **300** (i.e., the target object **30** does not appear in the user image captured by the terminal device **300**). Accordingly, the processor **110** may generate a pause command, in which the pause command may instruct the terminal device **300** to pause playing the demonstration image. The processor **110** may calculate an angle $\theta_{\text{sub.hand}}$ and an angle $\theta_{\text{sub.leg}}$ according to formula (1) and the formula (2), and may determine whether the angle $\theta_{\text{sub.hand}}$ or the angle $\theta_{\text{sub.leg}}$ is outside the default range according to formula (3), where $(x_{\text{sub.1}}, y_{\text{sub.1}})$ are the coordinates of the joint **31**, $(x_{\text{sub.2}}, y_{\text{sub.2}})$ are the coordinates of the joint **32**, $(x_{\text{sub.3}}, y_{\text{sub.3}})$ are the coordinates of the joint **33**, and $(x_{\text{sub.4}}, y_{\text{sub.4}})$ are the coordinates of the joint **34**. If formula (3) does not hold, the processor **110** may generate a suspend command. The default range of 10 to 180 degrees in formula (3) is determined based on multiple image samples of limb joints of different people, for example.

$$[00001] \quad \theta_{\text{hand}} = \cos^{-1} \left(\frac{(x_2 - x_1) \cdot \text{Math.}(x_4 - x_3) + (y_2 - y_1) \cdot \text{Math.}(y_4 - y_3)}{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \cdot \text{Math.}\sqrt{(x_4 - x_3)^2 + (y_4 - y_3)^2}} \right) \quad (1)$$

$$\theta_{\text{leg}} = \cos^{-1} \left(\frac{(x_3 - x_2) \cdot \text{Math.}(x_4 - x_1) + (y_3 - y_2) \cdot \text{Math.}(y_4 - y_1)}{\sqrt{(x_3 - x_2)^2 + (y_3 - y_2)^2} \cdot \text{Math.}\sqrt{(x_4 - x_1)^2 + (y_4 - y_1)^2}} \right) \quad (2) \quad 10^\circ \leq \theta_{\text{hand}}, \theta_{\text{leg}} \leq 108^\circ \quad (3)$$

[0061] The processor **110** may determine whether to generate a playback control command for playback speed control (e.g., fast forward, rewind, or slow motion) according to the expression of

the target object **30** (e.g., the user). Specifically, the processor **110** may perform facial recognition on a face **35** of the target object **30** (e.g., the user) in the user image to obtain multiple joints, such as a joint **41** representing the left eye, a joint **42** representing the right eye, a joint **43** representing the left corner of the mouth, and a joint **44** representing the right corner of the mouth. The processor **110** may calculate an area change of a polygon **40** formed by multiple joints. If the time when an absolute value of the area change is greater than a change threshold is greater than a time threshold, it means that the target object **30** (e.g., the user) may be surprised or confused by the demonstration image. Accordingly, the processor **110** may generate a playback control command (e.g., a slow-motion command) for reducing the playback speed. If the absolute value of the area change is less than or equal to the change threshold, or the time when the absolute value of the area change is greater than the change threshold is less than or equal to the time threshold, the processor **110** may not generate a playback control command for playback speed control. The processor **110** may calculate an area P_t of the polygon **40** at time point t according to formula (4), and may calculate an area change ΔP of the polygon **40** during a period $(t_{sub.2} - t_{sub.1})$ according to formula (5), where $(x_{sub.1}, y_{sub.1})$ represents the coordinates of the joint **31**, $(x_{sub.2}, y_{sub.2})$ represents the coordinates of the joint **32**, $(x_{sub.3}, y_{sub.3})$ represents the coordinates of the joint **33**, and $(x_{sub.4}, y_{sub.4})$ represents the coordinates of the joint **34**.

$$[00002] \quad P_t = \frac{1}{2} \cdot \text{Math.} \cdot x_1 y_2 + x_2 y_3 + x_3 y_4 + x_4 y_1 - x_2 y_1 - x_3 y_2 - x_4 y_3 - x_1 y_4 \cdot \text{Math.} \quad (4)$$

$$P = P_{t2} - P_{t1} \quad (5)$$

[0062] In an embodiment, the processor **110** may determine whether the face of the target object **30** (e.g., the user) is close to or far away from the playback device (e.g., the terminal device **300**) that plays the demonstration image according to the area change ΔP of the polygon **40**. If the area change ΔP is greater than the threshold, it means that the target object **30** (e.g., the user) may not be able to see the demonstration image clearly and the target object **30** (e.g., the user) has to move his/her face closer to the playback device. Accordingly, the processor **110** may generate a playback control command (e.g., a slow-motion command) for reducing the playback speed, so that the user can more easily see the details of the demonstration video. If the area change ΔP is less than or equal to the threshold, it means that the target object **30** (e.g., the user) can see the demonstration image clearly and therefore does not move his/her face closer to the playback device. Accordingly, the processor **110** may generate a playback control command for increasing (or resuming) the playback speed.

[0063] In an embodiment, the processor **110** may perform expression recognition on the face **35** of the target object **30** (e.g., the user) in the user image based on, for example, machine learning technology, and generate a playback control command for playback speed control (e.g., fast forward, rewind, or slow motion) according to a recognition result of the expression recognition. For example, if the recognition result of the expression recognition indicates that the user is confused, the processor **110** may generate a playback control command (e.g., a slow-motion command) for reducing the playback speed, so that the user can more easily see the details of the demonstration video.

[0064] Referring to FIG. 2, in an embodiment, after generating an adjustment command or a playback control command, the processor **110** may generate a script corresponding to the demonstration image. The script may include the adjustment command or the playback control command, and may include a timestamp corresponding to the adjustment command or the playback control command. The processor **110** may upload the script to the cloud server **500** for other users to download. For example, the terminal device **400** may download the script from the cloud server **500**. The terminal device **400** may execute the script when playing the demonstration image to adjust the display region or playback mode of the demonstration image. Accordingly, the terminal device **400** may automatically adjust the playback of the demonstration image without data transmission with the electronic device **100**.

[0065] FIG. 7 illustrates a flow chart of a method for adjusting a display region based on motion according to an embodiment of the disclosure. The method can be implemented by the electronic device **100** shown in FIG. 1. In step **S701**, a first image including a first target object is obtained. In step **S702**, the first image is detected to obtain a first joint of the first target object. In step **S703**, a first score is calculated according to a first displacement of the first joint. In step **S704**, in response to the first score being greater than a threshold, an adjustment command for adjusting a first display region of the first image is generated according to the first joint. In step **S705**, the adjustment command is output.

[0066] To sum up, the electronic device of the disclosure may determine which region in the image is an important display region that the user pays attention to according to the motion of the person in the played image. The electronic device may further determine the important display region according to the audio of the image. After obtaining the important display region, the electronic device may output commands according to the important display region to adjust the playback method of the image played by the terminal device of the user. The disclosure provides the user with a convenient method of image playback, and the user does not need to manually operate the terminal device to change the display region of the image played by the terminal device.

[0067] It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed embodiments without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the disclosure covers modifications and variations provided that they fall within the scope of the following claims and their equivalents.

Claims

1. An electronic device for adjusting a display region based on motion, comprising: a transceiver, obtaining a first image comprising a first target object; and a processor, coupled to the transceiver, configured to: detect the first image to obtain a first joint of the first target object; calculate a first score according to a first displacement of the first joint; in response to the first score being greater than a threshold, generate an adjustment command for adjusting a first display region of the first image according to the first joint; and output the adjustment command through the transceiver.
2. The electronic device according to claim 1, wherein the processor is further configured to: calculate a second score according to a second displacement of a second joint of the first target object; and in response to the second score being greater than the threshold, generate the adjustment command according to the first joint and the second joint.
3. The electronic device according to claim 2, wherein the processor is further configured to: divide a region of interest of the first image to obtain a plurality of sub-regions, wherein the sub-regions comprise a first sub-region and a second sub-region; determine that the first joint is located in the first sub-region and the second joint is located in the second sub-region according to the first image; and in response to the second sub-region being adjacent to the first sub-region, generate the adjustment command according to the first joint and the second joint.
4. The electronic device according to claim 3, wherein the sub-regions comprise a third sub-region corresponding to a third joint of the first target object, wherein a third score corresponding to the third joint is greater than the threshold, and the processor is further configured to: in response to the first sub-region being adjacent to the second sub-region and the third sub-region being adjacent to at least one of the first sub-region and the second sub-region, generate the adjustment command according to the first joint, the second joint, and the third joint.
5. The electronic device according to claim 3, wherein the processor is further configured to: generate the first display region according to at least one sub-region comprising the first sub-region; and in response to the second sub-region not being adjacent to the at least one sub-region, generate a second display region of the first image according to the second sub-region.
6. The electronic device according to claim 5, wherein the adjustment command instructs that the

first display region of the first image is output in a first period and the second display region of the first image is output in a second period.

7. The electronic device according to claim 6, wherein the processor is further configured to: in response to a size of the second display region being different from a size of the first display region, generate the adjustment command comprising a zoom in operation or a zoom out operation.

8. The electronic device according to claim 3, wherein the processor is further configured to: determine a boundary of the first display region according to a union of the first sub-region and the second sub-region.

9. The electronic device according to claim 1, wherein the processor is further configured to: obtain a first audio file corresponding to the first image through the transceiver; determine a first correlation between the first audio file and the first joint; and calculate the first score according to the first displacement and the first correlation.

10. The electronic device according to claim 9, wherein the processor is further configured to: perform a speech-to-text conversion on the first audio file to generate text; determine a number of words in the text associated with the first joint; and determine the first correlation according to the number of the words.

11. The electronic device according to claim 9, wherein the processor is further configured to: determine a first weight of the first displacement and a second weight of the first correlation according to a category of the first image; and calculate the first score according to the first displacement, the first weight, the first correlation, and the second weight.

12. The electronic device according to claim 1, wherein the processor is further configured to: obtain a second image comprising a second target object according to the transceiver; and generate a playback control command for the first image according to the second image, and output the playback control command through the transceiver.

13. The electronic device according to claim 12, wherein the processor is further configured to: detect the second image to obtain a plurality of limb joints of the second target object, wherein the limb joints comprise a left elbow joint, a right elbow joint, a left knee joint, and a right knee joint; calculate an angle according to the limb joints, and determine that the angle is within a default range; and in response to the angle being outside the default range, generate the playback control command for pausing the first image.

14. The electronic device according to claim 1, wherein the processor is further configured to: perform facial recognition on a second target object in a second image to obtain a plurality of joints; calculate an area change of a polygon formed by the joints; and in response to an absolute value of the area change being greater than a change threshold, generate a playback control command for playback speed control.

15. The electronic device according to claim 14, wherein the playback control command is used to reduce a playback speed of the first image.

16. The electronic device according to claim 1, wherein the processor is further configured to: perform facial recognition on a second target object in a second image to obtain a plurality of joints; calculate an area change of a polygon formed by the joints; and generate a playback control command for playback speed control according to the area change.

17. The electronic device according to claim 16, wherein the processor is further configured to: in response to the area change being greater than a first threshold, generate the playback control command for reducing a playback speed of the first image; and in response to the area change being less than or equal to a second threshold, generate the playback control command for increasing the playback speed of the first image.

18. The electronic device according to claim 1, wherein the processor is further configured to: detect the first image to obtain a center point and a height of the first target object; and determine a region of interest of the first image according to the center point and the height.

19. The electronic device according to claim 1, wherein the processor is further configured to:

output a script through the transceiver, wherein the script comprises the adjustment command and a timestamp of the first image corresponding to the adjustment command.

20. A method for adjusting a display region based on motion, comprising: obtaining a first image comprising a first target object; detecting the first image to obtain a first joint of the first target object; calculating a first score according to a first displacement of the first joint; in response to the first score being greater than a threshold, generating an adjustment command for adjusting a first display region of the first image according to the first joint; and outputting the adjustment command.
